Lab Exercise 12

Task 1: Student Record Management Using Structure

Objective

To apply the concept of **structure (`struct`)** in C programming by passing a structure **by value** to user-defined functions for data input and output.

Problem Statement

Write a C program to store and display a student's information using a **structure** and **functions**.

Program Requirements

1. Define a structure named `Student` with the following members:

```
struct Student {
    int id;
    char name[30];
    float cgpa;
}stdInfo;
```

2. Create the following user-defined functions:
   - `void readStudent()`
     → This function reads the student's data and copy all input to new global struct;
   - `void displayStudent(struct Student s)`
     → This function displays the student's information.
3. In the `main()` function:
   - Declare a variable of type `struct Student`
   - Call `readStudent()` to input the student details
   - Call `displayStudent()` to display the student record
4. Use sentinel loop to stop the looping

Sample Input/Output

```
Enter student details

Student ID: 1111
Name      : BAHARUDIN OSMAN
CGPA      : 3.5

--- Student Record ---
```

```
Student ID: 1111
Name       : BAHARUDIN OSMAN
CGPA       : 3.50

Any more data [Y/N] : Y

Enter student details

Student ID: 2222
Name       : AZMI OSMAN
CGPA       : 3.3

--- Student Record ---
Student ID: 2222
Name       : AZMI OSMAN
CGPA       : 3.30

Any more data [Y/N] : N
```

Task 2

You are given the following structure definition:

```
struct Employee {
    int empID;
    char name[30];
    float salary;
} emp[50];
```

Write a C program to store employee information and calculate the average salary using **functions**.

Program Requirements

1. Use the given structure `Employee`.
2. Declare a global array of structures to store max 50 employee records.
3. Implement the following functions:
   - `void inputEmployee()`
     → Reads employee ID, name, and salary.
   - `float calculateAverageSalary()`
     → Calculates and returns the average salary.
   - `void displayEmployee()`
     → Displays all employee records.
4. In the `main()` function:
   - Use sentinel loop to stop the looping
   - Call `inputEmployee()` to read data
   - Call `calculateAverageSalary()` and display the average salary
   - Call `displayEmployee()` to display employee details

Sample Input/Output

```
Enter details for employee 1
Employee ID: 11111
Name: BAHARUDIN
Salary: 4567
Anymore data [Y/N] : Y

Enter details for employee 2
Employee ID: 22222
Name: NIDURAHAB
Salary: 4345
Anymore data [Y/N] : Y

Enter details for employee 3
Employee ID: 33333
Name: SIMANANUS
Salary: 3456
Anymore data [Y/N] : Y

Enter details for employee 4
Employee ID: 44444
Name: TORAIMASU
Salary: 5674
Anymore data [Y/N] : N

Average Salary: RM 4510.50

Employee Records
ID       Name            Salary(RM)
===============================
11111    BAHARUDIN        4567.00
22222    NIDURAHAB        4345.00
33333    SIMANANUS        3456.00
44444    TORAIMASU        5674.00
```